Jithesh Pavan DSouza

Indian Institute of Science Education and Research, TVM, Kerala, India

jithesh1602@gmail.com GitHub Website

EDUCATION

IISER Thiruvananthapuram, Kerala | BS-MS Dual Degree Aug 2019– July 2024
Major Subject – Physics, Minor Subject – Data Science | 1st in Codebattle 2023

S.A.P.H.S.S School, Kerala | Higher Secondary School (State) 2017–2019
Subjects – Physics, Chemistry, Mathematics, Biology, English | Marks – 95.9%

Work Experience

AI Safety Research Assistant July 2025 – July 2026
Company: **AIPlans**
Model Diffing Project — Sole Implementer Code — Models

- Independently developed and ran model post-training and evaluation workflows, covering dataset preparation, fine-tuning, benchmarking, and checkpoint publication.
- Applied GRPO, PPO, DPO, and KTO to produce language-model variants for studying changes in model behavior; fine-tuned Qwen models at 0.6B and 8B scales across project experiments.
- Evaluated DPO-trained Qwen3-0.6B against its starting model using `lm-eval-harness` on ARC, HellaSwag, TruthfulQA, and WinoGrande.
- Curated preference-training datasets and published model checkpoints, training details, and evaluation results on Hugging Face.

Dobbe.ai Intern, Mar-June 2025, Delhi

- Built object detection and semantic segmentation pipelines (YOLO, SAM, Faster R-CNN, UNetResNet50) to segment and detect teeth parts and cavities, that were used in production.
- Came up with and built a fast, temporary method to segment cavities using k-means clustering
- *Tools:* PyTorch, Detectron2, Ultralytics, OpenCV

ExpoSys Datalabs Intern, Jun–Jul 2024, Bangalore

- Developed ML models (linear/logistic regression, neural nets) for regression tasks.

Research Projects

Mechanistic Interpretability Research Sept 2026 –Present
FutureProbes: Early Detection of LLM Misalignment
Advisor: Dr. Shivam Raval

- Developing lookahead linear probes to investigate whether LLM internal activations predict misaligned behavior before it appears in generated outputs.
- Implemented and tested a preliminary experimental framework for behavior-lookahead probing, including activation extraction and a linear-probe pipeline.

- Built experiments targeting reward hacking, sandbagging, manipulation, and deception to study early signals of potentially harmful model behavior.

Quantum ML for LLM-Based Healthcare Data August 2024– Dec 2024
Advisor: Dr. Ahmed Farooq

- Did sentiment and disease classification using QNNs and QKE with SVM and spectral clustering. Was able to achieve close to 90% accuracy.
- Explored $UU^{\dagger}$ encoding, GANs, varied tokenization/embedding strategies, and modeled quantum noise impacts. Manuscript in preparation: *Evaluating QML Algorithms for LLMs in Healthcare.*

IISER TVM – Major Thesis Aug 2023 – Apr 2024
Title: Neural Network Quantum States on Fermionic Systems Advisor: Dr. Amal Medhi

- Was successfully able to model complex strongly correlated 2D fermions hubbard systems using Machine Learning models.
- The model was built on Neural Network Quantum States (NQS) and was coupled with Monte Carlo for state filtering. I was able to train this complex interconnected system using Variational Monte Carlo (VMC) and Stochastic Reconfiguration (SR).
- I was able to simulate a 10×10 2D Hubbard lattice and observed a metal–insulator transition.
- **Modelling:** The NQS was built on a structured probabilistic graphical model- Restricted Boltzmann Machine (RBM) and Convoluted RBM (CRBM), to capture correlated variable interactions.
- **Optimization:** Faced gradient/SR instability and was able to fix the problem using linear algebra. We reduced the rank of the gradient matrix using Singular Value Decomposition (SVD) to remove noise.

Data Science Minor Project- IISER TVM Jan 2023 - April 2023
Title- **Deep Learning In Image Classification** Advisor: Prof. Nagaiah Chamakuri,

- Built and trained various Computer Vision model(Variants of AlexNet and VGG) to get high score on various datasets such as CIFAR.
- Did an in-depth study of the fundamentals of Machine Learning and Deep Learning.
- Understood Deep Learning, Neural Networks, Mathematical Optimization, & Image Processing.
- **Programming Languages :** PyTorch library from Python | Sklearn | Tensorflow etc..

Personal Research Projects

- My RAG Augmented chatbot – AI Alignment Chatbot
- Implemented a Transformer from scratch using PyTorch – link
- Built a program to generate artistic visuals using the deep dream algorithm – link
- Developing a browser-based application with AI-powered content filtering.

ACCOMPLISHMENTS & ACTIVITIES

- First prize in Codebattle 2023 Anvesha, IISER TVM
- My Blogs- link
- Scored 104 out of 120 in TOEFL iBT 2023 (English Proficiency exam)

Publications

- Evaluating Quantum Machine Learning Algorithms for Large Language Models in Healthcare Data. (Not Published yet) Draft